

Measurement (Term 2)

Choose the Appropriate Answer

1. The longest chord of a circle is called the
(a) Radius (b) Circumference (c) Diameter (d) Arc
Ans: (c) Diameter
2. The distance from the centre of a circle to any point on the circle is called the
(a) Diameter (b) Radius (c) Chord (d) Secant
Ans: (b) Radius
3. A line that intersects the circle at exactly one point is a
a. Secant (b) Chord (c) Diameter (d) Tangent
Ans: (d) Tangent
4. A part of the circumference of a circle is called an:
a. Sector (b) Segment (c) Arc (d) None of the above
Ans: (c) Arc
5. A circle can have infinitely many
(a) Radii (b) Diameters (c) Tangents (d) Chords
Ans: (c) Tangents
6. Formula used to find the circumference of a circle is
(a) $2\pi r$ units (b) $\pi r^2 + 2r$ units (c) πr^2 sq. units (d) πr^3 cu. units
Ans: $2\pi r$ units
7. In the formula, $C = 2\pi r$, 'r' refers to
(a) Circumference (b) Area (c) Rotation (d) Radius
Ans: (d) Radius
8. If the circumference of a circle is 82π , then the value of 'r' is
(a) 41 cm (b) 82 cm (c) 21 cm (d) 20 cm
Ans: (a) 41 cm
9. Circumference of a circle is always
(a) Three times of its diameter (b) More than three times of its diameter
(c) Less than three times of its diameter (d) Three times of its radius
Ans: (b) More than three times of its diameter
10. The formula used to find the area of the circle is sq. Units.
(a) $4\pi r^2$ (b) πr^2 (c) $2\pi r^2$ (d) $\pi r^2 + 2r$
Ans: (b) πr^2
11. The ratio of the area of a circle to the area of its semicircle is
(a) 2:1 (b) 1:2 (c) 4:1 (d) 1:4
Ans: (a) 2: 1

12. Area of a circle of radius 'n' units is
 (a) $2\pi r^p$ sq. units (b) πm^2 sq. units (c) πr^2 sq. units (d) πn^2 sq. units
Ans: (d) πn^2 sq. units

13. The formula to find the area of the circular path is
 (a) $\pi (R^2 - r^2)$ sq. units (b) πr^2 sq. units
 (c) $2\pi r^2$ sq. units (d) $\pi r^2 + 2r$ sq. units
Ans: (a) $\pi (R^2 - r^2)$ sq. units

14. The formula used to find the area of the rectangular path is
 (a) $\pi (R^2 - r^2)$ sq. units (b) $(L \times B) - (l \times b)$ sq. units
 (c) LB sq. units (d) lb sq. units
Ans: (b) $(L \times B) - (l \times b)$ sq. units

15. The formula to find the width of the circular path is
 (a) $(L - l)$ units (b) $(B - b)$ units (c) $(R - r)$ units (d) $(r - R)$ units
Ans: (c) $(R - r)$ units

16. In the formula, $C = 2\pi r$, 'r' refers to
 (a) Radius (b) Diameter (c) Circumference (d) None
Ans: (a) Radius

17. If the circumference of a circle is 82π , then the value of 'r' is
 (a) 44 cm (b) 41 cm (c) 43 cm (d) 42 cm
Ans: (b) 41 cm

18. Circumference of a circle is always
 (a) A line (b) a curve (c) π times of its diameter (d) Both (b) & (c)
Ans: (d) Both (b) & (c)

19. The radius of a circle is 7 cm. Its circumference is
 (a) 22 cm (b) 44 cm (c) 11 cm (d) 66 cm
Ans: (b) 44 cm

20. The diameter of a circle is 28 cm. Find its circumference
 (a) 14 cm (b) 24 cm (c) 88 cm (d) 66 cm
Ans: (c) 88 cm

21. The radius of a circle is 7 cm. Find its area
 (a) 154 cm^2 (b) 77 cm^2 (c) 11 cm^2 (d) 22 cm^2
Ans: (a) 154 cm^2

22. When the circumference and area of a circle are numerically equal, what is the diameter numerically equal to?
 (a) Area (b) Circumference (c) 271 (d) 4
Ans: (d) 4

Circumference of circle = Area of circle $2\pi r = \pi r^2 \quad \Rightarrow \quad d = \left(\frac{d}{2}\right)^2 \quad \Rightarrow \quad 4d = d^2 \quad \Rightarrow \quad d = 4$
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23. Find the area of a veranda 2.25 m wide constructed outside a room 5.5 m long and 4 m wide.

- (a) 36sq.m (b) 63sq.m (c) 64sq.m (d) 84sq.m

Ans: (d) 84sq.m

24. The distance around the circular region is called _____.

- (a) Area (b) Circumference (c) Line (d) None of these

Ans: (b) Circumference

25. _____ is the part of plane occupied by the closed figure.

- (a) Area (b) Circumference (c) Line (d) None of these

Ans: (a) Area

26. What is the circumference of the circle diameter is 10 cm?

- (a) 35 cm (b) 30 cm (c) 31.4 cm (d) None of these

Ans: 31.4 cm

27. Find the radius of the circle 9.8 m.

- (a) 4.9 m (b) 19.6 m (c) 10 m (d) None of these

Ans: (a) 4.9 m

28. If the radius of pipe is 1 cm , the circumference of pipe will be

- (a) 62.8 cm (b) 6.28 cm (c) 62.8 cm² (d) 6.28 cm

Ans: (b) 6.28 cm

29. The circumference of the circle is

- (a) πr (b) πr^2 (c) $\pi \times 2r$ (d) $\pi + 2r$

Ans: (c) $\pi \times 2r$

30. The diameter of a circle is

- (a) r^2 (b) $2r$ (c) $2\pi r^3$ (d) πr^2

Ans: (b) $2r$

31. Which of the following is an example of circle?

- (a) a chair (b) a bottle cap (c) a cup (d) a table

Ans: (b) a bottle cap

32. The area of a circle is

- (a) $2\pi r$ (b) $2\pi r^2$ (c) πr^2 (d) πd

Ans: (c) πr^2

33. $1 \text{ m}^2 =$ _____.

- (a) 100 c m² (b) 1000 c m² (c) 10000 m² (d) 10000 c m²

Ans: (d) 10000 c m²

34. One hectare is equal to

- (a) 100 m² (b) 1000 m² (c) 10,000 m² (d) 10,000 m

Ans: (c) 10,000 m²

35. The circumference of a circle with radius 7 cm is
 (a) 11 cm (b) 22 cm (c) 44 cm (d) 49 cm
Ans: (c) 44 cm

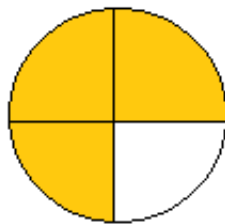
36. The perimeter of circle is its
 (a) Area (b) Circumference (c) Radius (d) Diameter
Ans: (b) Circumference

37. Diameter is _____.
 (a) Twice radius (b) Half radius (c) Equal to radius (d) One-third of radius
Ans: (a) Twice radius

38. π (pi) is
 (a) Ratio of circumference to diameter (b) 21 / 17
 (c) Diameter to circumference (d) 3.14
Ans: (a) Ratio of circumference to diameter

39. The value of constant π is
 (a) 31.4 (b) 22 / 7 (c) 7 / 22 (d) 314
Ans: (b) 22 / 7

40. If the area of circle is 44 cm^2 , the area of shaded portion will be



- (a) 11 cm^2 (b) 11 cm (c) 22 cm (d) 33 cm^2
Ans: (d) 33 cm^2

41. The area of a circle is $49\pi \text{ cm}^2$. Its circumference is
 (a) $7\pi \text{ cm}$ (b) $14\pi \text{ cm}$ (c) $21\pi \text{ cm}$ (d) $28\pi \text{ cm}$
Ans: (b) $14\pi \text{ cm}$

Area of the circle $= \pi r^2 = 49\pi \Rightarrow r = 7 \text{ cm}$ Circumference of the circle $= 2\pi r = 14\pi \text{ cm}$
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42. The area of a circle is 38.5 cm^2 . Its circumference is
 (a) 62 cm (b) 12.1 cm (c) 11 cm (d) 22 cm
Ans: (d) 22 cm

The area of a circle $A = \pi r^2 = 38.5$ $r^2 = \frac{38.5 \times 7}{22} = \frac{269.5}{22} = 12.25 \Rightarrow r = \sqrt{12.25} = 3.5 \text{ cm} = \frac{7}{2} \text{ cm}$ The circumference of a circle $C = 2\pi r \Rightarrow 2 \times \frac{22}{7} \times \frac{7}{2} = 22 \text{ cm}$
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43. The circumference of two circles are in the ratio 2 : 3. The ratio of their areas is
 (a) 2 : 3 (b) 4 : 9 (c) 9 : 4 (d) None of these

Ans: (b) 4: 9

$$\text{The circumference of a circle is given by } C = 2\pi r.$$

$$\frac{2\pi r_1}{2\pi r_2} = \frac{2}{3}, \text{ which simplifies to } \frac{r_1}{r_2} = \frac{2}{3} \Rightarrow \left(\frac{r_1}{r_2}\right)^2 = \frac{4}{9}$$

44. The areas of the two circles are in the ratio 4 : 9. The ratio of their circumference is
 (a) 2 : 3 (b) 4 : 9 (c) 9 : 4 (d) 4 : 9

Ans: (a) 2: 3

45. Find the circumference of a circle of diameter 21 cm.
 (a) 62 cm (b) 64 cm (c) 66 cm (d) 68 cm

Ans: (c) 66 cm

$$\text{Circumference of the circle} = C = \frac{22}{7} \times 21 \text{ cm}$$

$$C = 22 \times 3 \text{ cm} = 66 \text{ cm.}$$

46. Find the area of a circle whose circumference is 52.8 cm.
 (a) 221.76 cm² (b) 220.76 cm² (c) 200.76 cm² (d) None of these.

Ans: (a) 221.76 cm²

$$r = \frac{52.8}{2 \times \frac{22}{7}} \text{ cm} = \frac{52.8 \times 7}{44} \text{ cm} = \frac{369.6}{44} \text{ cm} = 8.4 \text{ cm.}$$

$$A = \frac{22}{7} \times (8.4)^2 \text{ cm}^2 = \frac{22}{7} \times 70.56 \text{ cm}^2 = 22 \times 10.08 \text{ cm}^2 = 221.76 \text{ cm}^2.$$

47. The circumference of a circle is 44 cm. Find the radius of the circle.
 (a) 7 cm (b) 14 cm (c) 15 cm (d) 21 cm

Ans: (a) 7 cm

$$\text{The circumference of the circle } C = 2\pi r = 44 \text{ cm}$$

$$r = \frac{C}{2\pi} = \frac{44 \text{ cm}}{2 \times \frac{22}{7}} = \frac{44 \times 7}{44} \text{ cm} = 7 \text{ cm}$$

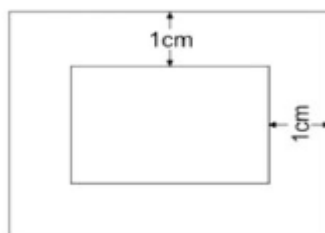
48. Find the circumference of a circle 220 cm. Find the diameter of the circle.
 (a) 60 cm (b) 70 cm (c) 66 cm (d) 68 cm

Ans: (b) 70 cm

$$\text{Circumference of circle } 220 \text{ cm} = \frac{22}{7} \times d.$$

$$d = \frac{220 \times 7}{22} \text{ cm} = 10 \times 7 \text{ cm} = 70 \text{ cm.}$$

49. In the figure, the inner rectangle is 1 cm inside the outer one. The outer rectangle has the perimeter of 26 cm. what is the perimeter of the inner one?



- (a) 24 cm (b) 22 cm (c) 20 cm (d) 18 cm

Ans: (b) 22 cm

Outer rectangle is 26 cm. Difference in outer and inner rectangle : 1 cm
 Perimeter = $1 \times 4 = 4 = 26 - 4 = 22$ cm.

50. The diameter of the wheels of a bus is 140 cm. Find the radius of the wheel.

- (a) 80 cm (b) 70 cm (c) 90 cm (d) 60 cm

Ans: (b) 70 cm

$$r = \frac{140}{2} = 70 \text{ cm}$$

51. The radius of the circle is 30 cm, Find its diameter.

- (a) 15 cm (b) 60 cm (c) 45cm (d) None of these

Ans: (b) 60 cm

$$\text{Diameter} = 2 \times 30 \text{ cm} = 60 \text{ cm}$$

52. The value of π is equal to

- (a) Area / Circumference (b) Area / Diameter
 (c) Circumference / Diameter (d) Circumference / Radius

Ans: (c) Circumference / Diameter

53. The perimeter of the semicircle which radius is 7 is

- (a) 27 cm (b) 36 cm (c) 26 cm (d) 40 cm

Ans: (b) 36 cm

$$\text{Perimeter of the semicircle} = \pi r + 2r = 22 + 14 = 36 \text{ cm}$$

54. The diameter of a circle of circumference 66cm.

- (a) 10.5cm (b) 14cm (c) 21cm (d) 7cm

Ans: (c) 21cm

$$d = \frac{66}{\frac{22}{7}} = 66 \times \frac{7}{22} = 3 \times 7 = 21 \text{ cm}$$

55. The area and circumference of a circle are numerically equal .Then radius is

- (a) 2 units (b) 4 units (c) 1 unit (d) 6 units

Ans: (a) 2 units

$$\pi r^2 = 2\pi r \Rightarrow r = 2 \text{ units}$$

56. Area of a circle with diameter (m), radius (n) and circumference (p) is

- (a) $2\pi n$ (b) πn^2 (c) πp^2 (d) πm^2

Ans: (b) πn^2

57. Circumference of a circle is always

- (a) More than three times of its diameter (b) Three times of its diameter
(c) Less than three times of its diameter (d) Three times of its radius

Ans: (a) More than three times of its diameter

58. Ratio of the circumference of a circle to its diameter is denoted by symbol_____.

- (a) C (b) D (c) π (d) R

Ans: (c) π

59. The circumference of a circle of diameter 10 cm is _____

- (a) 20π (b) 10π (c) 40π (d) None of these

Ans: (b) 10π

60. A school campus is rectangular in shape. Its length and breadth are 50m and 30m, respectively. There is a 2 m wide path inside the campus all around it. Find the area of the path in square metres?

- (a) 96 m^2 (b) 304 m^2 (c) 172 m^2 (d) 490 m^2

Ans: (b) 304 m^2

$$\begin{aligned} \text{The area of the entire campus} &= 50 \text{ m} \times 30 \text{ m} = 1500 \text{ m}^2. \\ \text{The area of the inner rectangle} &= 46 \text{ m} \times 26 \text{ m} = 1196 \text{ m}^2. \\ \text{The area of the path} &= 1500 \text{ m}^2 - 1196 \text{ m}^2 = 304 \text{ m}^2. \end{aligned}$$

61. A sheet of paper measures 30 cm by 20 cm. A strip of 4 cm wide is cut from it all around. Find the area of the remaining sheet?

- (a) 125 m^2 (b) 178 m^2 (c) 492 m^2 (d) 264 m^2

Ans: (d) 264 m^2

$$\begin{aligned} \text{The area of the original sheet of paper} &= 30 \text{ cm} \times 20 \text{ cm} = 600 \text{ cm}^2 \\ \text{The area of the remaining sheet} &= 22 \text{ cm} \times 12 \text{ cm} = 264 \text{ cm}^2 \end{aligned}$$

62. π is a

- (a) Repeating & terminating number (b) Non repeating & non-terminating number
(c) Rational number (d) none of these

Ans: (b) Non repeating & non-terminating number

63. The circumference of the bangle is 462 mm, the diameter is

- (a) 146 mm (b) 147 mm (c) 148 mm (d) 149 mm

Ans: (b) 147 mm

$$C = \pi d \Rightarrow 462 = \pi d$$

$$d = 462 \times \frac{7}{22} = 147 \text{ mm}$$

64. Area between two co - centric circles _____.

- (a) $A = \pi r$ (b) $A = \pi r^2$ (c) $A = \pi (R + r) (R - r)$ (d) $A = \pi r^2 h$

Ans: (c) $A = \pi (R + r) (R - r)$

65. Area of the circle is 154 cm^2 . Find its diameter.

- (a) 7 cm (b) 14 cm (c) 49 cm (d) None of these

Ans: (b) 14 cm

$$\text{The area of the circle} = \pi r^2 = 154 \text{ cm}^2.$$

$$r^2 = \frac{154 \times 7}{22} = 49 \Rightarrow r = 7 \text{ cm} \Rightarrow d = 2 \times 7 = 14 \text{ cm}$$

66. The circumference of the two circles are 256 cm and 512 cm, then the ratio of their circumference are _____.

- (a) 1:2 (b) 1:3 (c) 3: 2 (d) 2:3

Ans: (a) 1:2

67. 5m lawn is built along the outside of the rectangular park with measurement 40 m and 30 m then the area of the lawn is given by

- (a) 600 m^2 (b) 800 m^2 (c) 1000 m^2 (d) 1200 m^2

Ans: (b) 800 m^2

$$\begin{aligned} \text{Area of the inner rectangle} &= 40 \text{ m} \times 30 \text{ m} = 1200 \text{ m}^2 \\ \text{Area of the outer rectangle} &= 50 \text{ m} \times 40 \text{ m} = 2000 \text{ m}^2 \\ \text{Area of the lawn} &= 2000 \text{ m}^2 - 1200 \text{ m}^2 = 800 \text{ m}^2. \end{aligned}$$

68. If the radius of the circle $\pi / 8 \text{ cm}$, The its circumference is given by,

- (a) $\pi / 4 \text{ cm}$ (b) $2\pi / 16 \text{ cm}$ (c) $\pi^2 \text{ cm}$ (d) $2\pi^2 \text{ cm}$

Ans: (a) $\pi / 4 \text{ cm}$

69. If the radius of the circle is doubled, then the area is

- (a) Doubled (b) Halved (c) 3 times (d) 4 times

Ans: (d) 4 times

70. A plastic sheet 12 cm long and 8 cm wide from it a circular sheet of radius 3.5 cm is cut, then the reminder area of sheet left is

- (a) 57.5 cm^2 (b) 58 cm^2 (c) 57.73 cm^2 (d) None of these

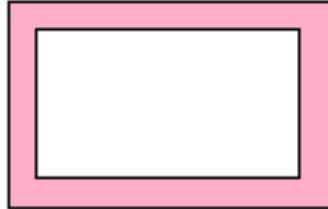
Ans: (a) 57.5 cm^2

$$\begin{aligned} A_{\text{rect}} &= 12 \times 8 = 96 \\ A_{\text{circ}} &= \pi (3.5)^2 = (22/7) \times (3.5) \times (3.5) = 38.5 \\ \text{Remaining area} &= A_{\text{rect}} - A_{\text{circ}} = 96 - 38.5 = 57.5 \end{aligned}$$

71. If the radius of the pipe is 1 cm, the circumference of the pipe will be
 (a) 6.28 cm (b) 62.8 cm² (c) 6.28 cm² (d) 62.8 cm

Ans: (a) 6.28 cm

72. In the figure the area of the larger rectangle is 1750 m² and the area of the smaller rectangle is 1350 m². Find the remaining area of the path between the rectangles.

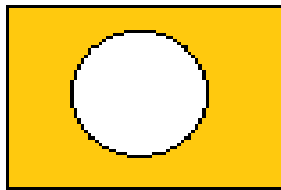


- (a) 3100 m² (b) 400 m² (c) 750 m² (d) 350 m²

Ans: (b) 400 m²

$$\text{Remaining Area} = 1750 \text{ m}^2 - 1350 \text{ m}^2 = 400 \text{ m}^2$$

73. In the figure the rectangular sheet is 50 cm² and the area of circle inside the sheet is 15 cm² cut from the sheet then the area of remaining sheet will be



- (a) 35 cm² (b) 35 cm (c) 65 cm² (d) 65 cm

Ans: (a) 35 cm²

$$\text{Remaining Area} = 50 \text{ cm}^2 - 15 \text{ cm}^2 = 35 \text{ cm}^2$$

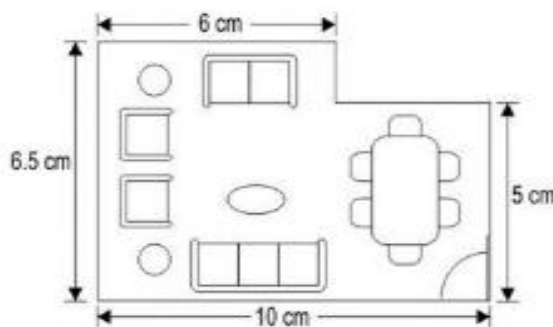
74. If r and d are the radius and diameter of a circle respectively, then
 (a) $d = 2r$ (b) $d = r$ (c) $d = 1/2 r$ (d) $d = r^2$

Ans: (a) $d = 2r$

75. 1 cm² =
 (a) 10 mm² (b) 100 mm² (c) 1000 mm² (d) 10000 mm²

Ans: (b) 100 mm²

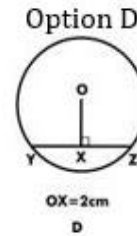
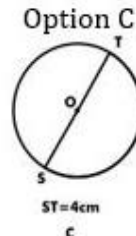
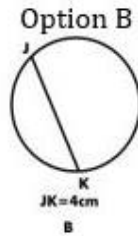
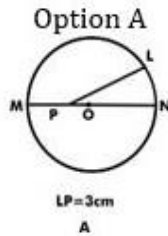
76. To find the perimeter of the room (in centimetres)?



- (a) 33 (b) 32.5 (c) 16.5 (d) 13.75

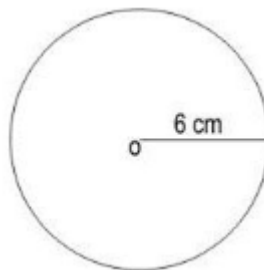
Ans: (c) 16.5

77. Which one of the circle below could have the radius 2 cm? (wherever mentioned O as the centre)



Ans: (c) Option c

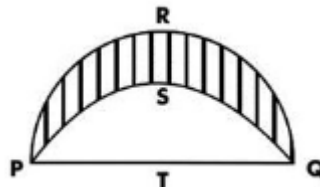
78. The largest square piece that can be cut out from a circular piece of paper of radius 6 cm is the one with.



- (a) Side 6 cm (b) Perimeter 12 cm (c) Diagonal 12 cm (d) Side 12 cm

Ans: (a) Side 6 cm

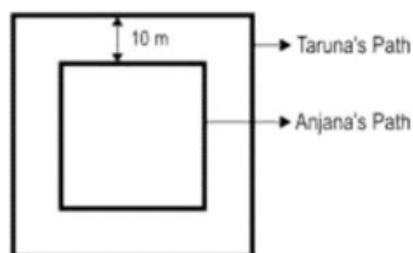
79. What is the correct way to find the perimeter of the shaded part of the given figure?



- (a) Length of arc PRQ – Length of arc PSQ
(b) Length of arc PRQ + Segment PQ - Length of arc PSQ
(c) Length of arc PRQ + Length of arc PSQ
(d) Length of arc PSQ + segment PQ

Ans: (c) Length of arc PRQ + Length of arc PSQ

80. Every morning Taruna covers the distance 1600 m. what is the distance covered by her sister Anjana?



- (a) 400 m (b) 1520 m (c) 1560 m (d) 1590 m

Ans: (b) 1520 m

81. The ratio of the radii of two circles are 3: 2. What are the circumferences?

- (a) 2: 3 (b) 1: 3 (c) 3: 2 (d) 3:1

Ans: (c) 3: 2

82. A circle has area which is 100 times the area of another circle. What is the area of their circumference?

- (a) 10: 1 (b) 1: 10 (c) 1: 1 (d) 2: 1

Ans: (a) 10: 1

83. The ratio of the area of a circle to the area of semicircle is

- (a) 1: 2 (b) 2: 1 (c) 1: 4 (d) 4: 1

Ans: (b) 2: 1

84. A circle is inscribed in a square of side 28 cm. Find the area of the circle.

- (a) 661 cm² (b) 616 cm² (c) 166 cm² (d) 660 cm²

Ans: (b) 616 cm²

$$A = \frac{22}{7} \times (14)^2 = \frac{22}{7} \times 196 = 22 \times 28 = 616 \text{ cm}^2$$

85. A piece of wire in the form of a rectangle with dimensions 12 m by 10 m is bent to form a circle. Find the diameter of the circle.

- (a) 41 m (b) 17 m (c) 14 m (d) 7 m

Ans: 14 m

The perimeter of the rectangle is $2 \times (12 + 10) = 2 \times 22 = 44 \text{ m}$.

The circumference of a circle $C = \pi d$

$$\Rightarrow 44 = \frac{22}{7} \times d \Rightarrow d = \frac{44 \times 7}{22} = 2 \times 7 = 14 \text{ m.}$$

86. Determine the radius of a circle whose circumference is 110 cm.

- (a) 15.5 cm (b) 71.5 cm (c) 51.5 cm (d) 17.5 cm

Ans: (d) 17.5 cm

$$C = 2\pi r \Rightarrow 110 = 2 \times \frac{22}{7} \times r.$$

$$\therefore r = \frac{110 \times 7}{44} = \frac{10 \times 7}{4} = \frac{70}{4} = 17.5 \text{ cm.}$$

87. Find the circumference of a circle with diameter 10 cm.

- (a) $31 \frac{3}{7}$ (b) $7 \frac{31}{3}$ (c) $3 \frac{7}{31}$ (d) $4 \frac{31}{7}$

Ans: $31 \frac{3}{7}$

$$C = \frac{22}{7} \times 10 \text{ cm} = \frac{220}{7} \text{ cm} = 31 \frac{3}{7}.$$

88. The area of the circle is doubled. What is the ratio of the new circle of the area of the given circle?

- (a) 1: 4 (b) 2: 1 (c) 1: 2 (d) 4: 1

Ans: 4: 1

89. Diameter of the circular garden is 9.8 cm. which of the following is its area?

- (a) 75.46 cm² (b) 76.46 cm² (c) 74.4 cm² (d) 76.4 cm²

Ans: (a) 75.46 cm²

$$A = \frac{22}{7} \times (4.9)^2 = \frac{22}{7} \times 4.9 \times 4.9 = 75.46 \text{ cm}^2$$

90. The angle subtended by a diameter at any point on the circle is

- (a) 90° (b) 180° (c) 60° (d) 45°

Ans: (a) 90°

This angle is always a right angle.

91. The diameter of the circle is

- (a) 2r (b) r (c) 3r (d) 4r

Ans: (a) 2r

92. A circular lawn has a diameter of 20 m. What is its circumference?

- (a) 20π cm (b) 40π cm (c) 10π cm (d) 100π cm

Ans: 20π cm

93. The radii of two circles are 19 cm and 9 cm respectively. The radius of the circle which has circumference equal to the sum of the circumference of two circles are

- (a) 27 cm (b) 28 cm (c) 26 cm (d) None of these

Ans: (b) 28 cm

$$c = 2\pi(19) + 2\pi(9) = 56\pi$$

$$2\pi r = 56\pi \Rightarrow r = 56\pi / 2\pi = 28 \text{ cm}$$

94. The radii of two circles are 8 cm and 6 cm respectively. The radius of the circle having to the sum of the areas of two circles are

- (a) 10 cm (b) 12 cm (c) 11 cm (d) None of these

Ans: (a) 10 cm

$$A_{\text{sum}} = \pi(8)^2 + \pi(6)^2 = 100\pi \text{ cm}^2$$

$$R^2 = 100 \Rightarrow R = \sqrt{100} = 10 \text{ cm}$$

95. If the perimeter and the area of a circle are numerically equal, then the diameter of the circle is

- (a) 4 units (b) 2π units (c) 8 units (d) 14 units

Ans:

$$2\pi r = \pi r^2 \Rightarrow r = 2 \Rightarrow d = 4$$

96. The area of the ring with radius of outer circle is 21 cm and the radius of the inner circle is 7 cm is calculated in terms of π .

- (a) 392π (b) 293π (c) 239π (d) None of these

Ans: (a) 392π

$$A_{\text{ring}} = A_{\text{outer}} - A_{\text{inner}} = \pi(21)^2 - \pi(7)^2 = 441\pi - 49\pi = 392\pi \text{ cm}^2$$

97. Part of circular region enclosed between a chord and the corresponding arc is called ----- of the circle.

- (a) Arc area (b) Segment (c) Sector (d) None of these

Ans: (b) Segment

98. The measure of an inscribed angle that intersects at semi-circle is

- (a) Acute angle (b) Right angle (c) Obtuse angle (d) None of these

Ans: (b) Right angle

99. The radius of the circle whose area is equal to the sum of the areas of the two circles with radius 3 cm and 4 cm is -----

- (a) 5 cm (b) 25 cm (c) 5π cm (d) None of these

Ans: (a) 5 cm

100. Which one of the following is true?

- (a) Circumference of the circle bears a constant ratio with its diameter.
(b) Circumference of the circle bears a constant ratio with its radius.
(c) Circumference of the circle bears a constant ratio with its area.
(d) None of these.

Ans: (a) Circumference of the circle bears a constant ratio with its diameter.